

From College to Career:

A Statistical Analysis of Student Employability Factors

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Introduction

In a competitive labor market, students' choices such as major, GPA, internships, and soft skill development shape early career outcomes. Yet advising is often based on intuition rather than quantitative evidence, which can lead to inefficient use of students' limited time. This gap is increasingly solvable with the availability of education and career datasets that enable evidence based evaluation of which factors matter most. This study asks a central question: which student investments have the greatest impact on career success? The answer has clear implications for prioritizing GPA, internships, and soft skills. Beyond individual guidance, these insights can help universities and career centers allocate resources more effectively and better support student outcomes.

Research Questions

This project examines which student factors are most strongly associated with early career outcomes, specifically starting salary, number of job offers, and time to promotion, using the *Education and Career Success* dataset. It focuses on three domains: academic performance (GPA and test scores), experiential learning (internships, projects, certifications), and soft skills (including networking).

The study is structured around three research questions:

RQ1: Do starting salary distributions differ across fields of study?

- Null Hypothesis (H_0) : The distribution of starting salaries is the same across all fields of study.
- Alternative Hypothesis (H_A) : At least one field of study has a different starting salary distribution.

RQ2: Is there a statistically significant difference in median starting salary between students with no internships and those with two or more internships?

- Null Hypothesis (H_0) : The median starting salary is the same for students with 0 internships and those with ≥ 2 internships.
- Alternative Hypothesis (H_A) : The median starting salary differs between these two groups.

RQ3: Which academic, experiential, and soft skill factors are most strongly associated with starting salary, job offers, and time to promotion?

Dataset

The dataset used in this study is the *Education and Career Success* dataset, publicly available through Kaggle. It is easily accessible as a single CSV file and contains 400 student records across 19 variables. The dataset includes five key categories of variables: demographics (age, gender), academic performance (high school GPA, SAT score, university GPA, field of study), experiential learning (internships, projects, certifications), soft skills (soft skills and networking scores), and career outcomes (starting salary, job offers, time to promotion, career satisfaction). This structure supports analysis of how different types of student attributes relate to early career success.

Prior to analysis, basic preprocessing steps are performed, including checking for missing values, removing duplicates, validating variable ranges, and grouping GPA into tiers to support comparison. These steps ensure consistency and reliability in the analysis. Although the dataset is synthetic, it is designed to reflect realistic relationships between education and career outcomes, making it suitable for exploratory and comparative analysis. Its structured nature and comprehensive variable coverage make it appropriate for identifying patterns and evaluating the relative importance of academic, experiential, and soft skill factors.

Methodology

To address the research questions, this study applies a set of complementary analytical methods chosen to capture both distributional differences and predictor importance. Differences in starting salary distributions across fields of study are first visualized using Kernel Density Estimation and then formally tested using the Kolmogorov Smirnov test, which provides a nonparametric assessment of whether two distributions differ.

For internship effects, bootstrap resampling is used to estimate confidence intervals for the median salary difference between students with no internships and those with two or more, allowing inference without assuming a specific distributional form.

To evaluate the relative importance of student factors, Pearson correlation analysis is used to measure linear relationships between variables and career outcomes, while a Random Forest model is used to capture nonlinear patterns and interactions among predictors. Feature importance scores from the Random Forest provide a ranking of variables based on their contribution when considered jointly. This combination of methods is deliberately chosen to provide a more complete and reliable analysis, since correlation alone cannot capture complex relationships and model based importance alone can obscure simple linear effects.

Results and Analysis

RQ1: Do Starting Salary Distributions Differ Across Fields of Study?

Kernel Density Estimation was used to visualize starting salary distributions across fields, revealing clear differences in both level and spread. High earning fields such as Computer Science, Engineering, and Medicine show much higher salary concentrations than lower earning fields such as Arts, Education, and Nursing.

To formally test these differences, pairwise Kolmogorov Smirnov tests were conducted across 55 field combinations. The null hypothesis was rejected in 45 out of 55 pairs (82%), with p-values below the 0.05 significance threshold. Several pairs produced KS statistics of exactly 1.0, indicating complete non-overlap between distributions. These perfect separations were concentrated between high earning fields (Computer Science, Engineering, Medicine) and low earning fields (Arts, Education, Nursing), confirming that these groups occupy entirely distinct salary ranges with no overlap.

The high earning cluster of Computer Science, Engineering, and Medicine showed significant differences even among themselves, indicating that while they share a higher salary range relative to other fields, they maintain distinct internal distributions. Notably, the uniformly low p-values and several KS statistics at 1.0 should be interpreted with caution given the synthetic nature of the dataset, which likely produces cleaner separations than would be observed in real world salary data.

The null hypothesis is rejected. Starting salary distributions differ significantly across fields of study, with field of study serving as a strong stratifying factor in early career earnings.

RQ2: Is there a statistically significant difference in median starting salary between students with no internships and those with two or more internships?

To assess the impact of internship experience on starting salary, students were divided into two groups: those who completed no internships and those who completed two or more. Bootstrap resampling with 10,000 iterations was used to estimate the 95% confidence interval around the observed median salary difference, avoiding assumptions about the underlying distribution.

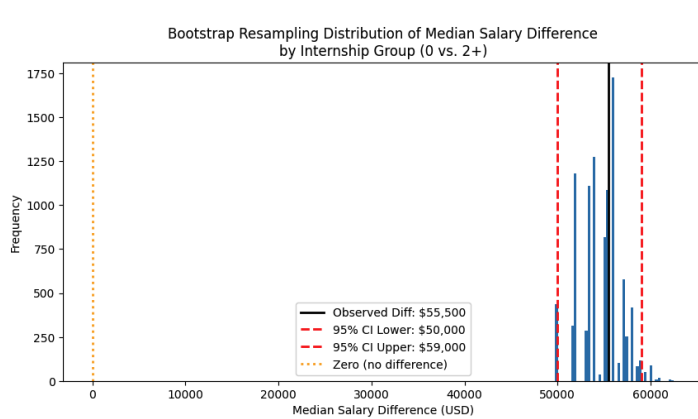


Figure 1. Bootstrap resampling distribution of median salary difference between students with 0 and 2+ internships ($n = 10,000$ iterations).

As shown in Figure 1, the bootstrap distribution is tightly concentrated between \$50,000 and \$60,000, with the 95% confidence interval [\$50,000, \$59,000] entirely above zero. Because the interval does not include zero, the null hypothesis is rejected. This indicates that students who complete two or more internships have a median starting salary approximately \$55,500 higher than those with no internships, and that this difference is statistically significant at the 95% confidence level.

RQ3: Which academic, experiential, and soft skill factors are most strongly associated with starting salary, job offers, and time to promotion?

Pearson correlation coefficients between the eight student factors and the three outcomes show uniformly strong associations across the board. Every factor correlates with starting salary at $r > 0.93$ and with job offers at $r > 0.95$, while all factors show strong negative correlations with years to promotion ($r < -0.92$), indicating that higher scores on any factor are associated with faster advancement.

In the correlation ranking for starting salary, Certifications lead at $r = 0.968$ and Internships_Completed last at 0.938. The notably narrow range of correlations (0.938–0.968) and the uniformly deep red heatmap pattern suggest that correlation alone does not meaningfully distinguish among predictors in this synthetic dataset, as all variables move together with outcomes in a highly structured and collinear manner.

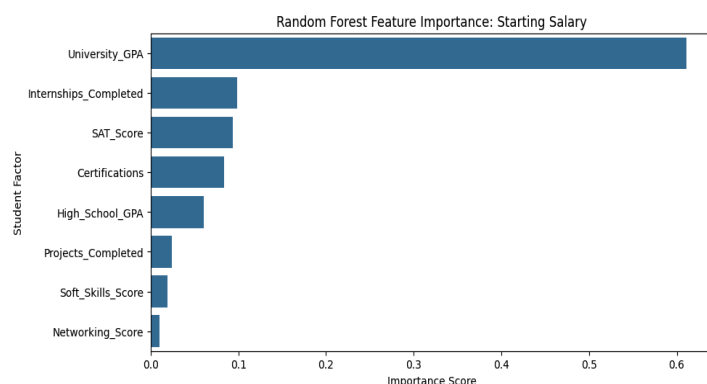


Figure 2. Random Forest feature importance scores for predicting starting salary across eight student factors.

As illustrated in the Random Forest model in Figure 2, feature importance rankings diverge sharply from the correlation results. University_GPA was the dominant predictor (~ 0.61), accounting for over 60% of total importance. In contrast, Soft_Skills_Score and Networking_Score, which were relatively strong in correlation, contributed little once other variables were considered. This reversal indicates that correlation overstated the role of some soft skill measures due to overlap between predictors, while the model identifies GPA and internships as the primary independent drivers of salary.

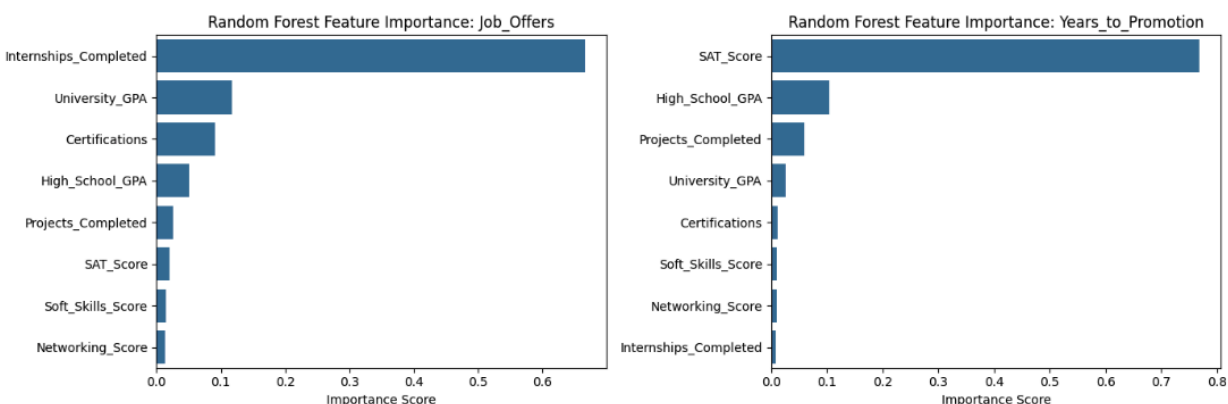


Figure 3. Random Forest feature importance scores for job offers (left) and years to promotion (right).

As shown in Figure 3 (left), Internships Completed dominates job offer attainment with an importance score of approximately 0.65, while all other variables contribute minimally. This strongly supports the central hypothesis that internship experience is the primary driver of job offers, outperforming academic and soft skill measures by a substantial margin.

For years to promotion, Figure 3 (right) shows a different pattern, with SAT Score emerging as the dominant predictor at approximately 0.77. This suggests that faster advancement is more closely associated with foundational cognitive ability than with experiential or collegiate performance. Notably, Internships Completed, which is most important for job offers, ranks last for promotion speed, reinforcing that the factors that secure employment are not necessarily the same factors that drive advancement once employed.

Discussion and Insights

Evidence from this analysis suggests a shift in career advising toward goal-specific guidance rather than a one-size-fits-all approach. Field of study functions as a structural ceiling on earning potential, making it the most consequential early decision. Beyond that, the results show a clear separation of roles among key predictors: GPA most strongly relates to starting salary, internships to job offers, and cognitive aptitude to

speed of advancement. These factors are not interchangeable, and treating them as such can lead to suboptimal advising. Certifications stand out as a flexible secondary investment, contributing to both salary and job outcomes with relatively lower time cost. Soft skills and networking, while valuable in practice, did not emerge as independent predictors in this analysis and should be positioned as complementary rather than primary strategic priorities.

Although the study provides a structured framework for evidence-based advising, its conclusions are limited by synthetic data and missing contextual variables. Future work should use real-world longitudinal datasets and incorporate factors such as geography, industry, and company size to improve external validity. Field-specific models would also help capture differences in predictor importance across disciplines, producing more precise and actionable insights than the pooled approach used here.

Use of AI Tools

Claude was used as a supplementary learning and writing support tool throughout the project. Specifically, it was used to explore and understand analytical methods that were not directly covered in class. In addition, Claude was used for writing assistance, particularly to refine clarity, structure, and academic tone in sections of the report. All analytical decisions, final interpretations, and conclusions were ultimately made independently.

Conclusion

This study aimed to replace intuition with evidence in the context of career advising, and the results provide a clear and consistent set of findings. Career success is not driven by a single universal factor, but by different combinations of predictors depending on the specific outcome being considered. Although the analysis is limited by the use of synthetic data, the framework is fully transferable to real-world datasets. It provides a practical model for how career centers and academic institutions can move toward more data-driven advising systems grounded in measurable relationships rather than intuition alone.

References

Shamim, A. (2025). *Education and Career Success* [Dataset]. Kaggle. <https://www.kaggle.com/datasets/adilshamim8/education-and-career-success>